

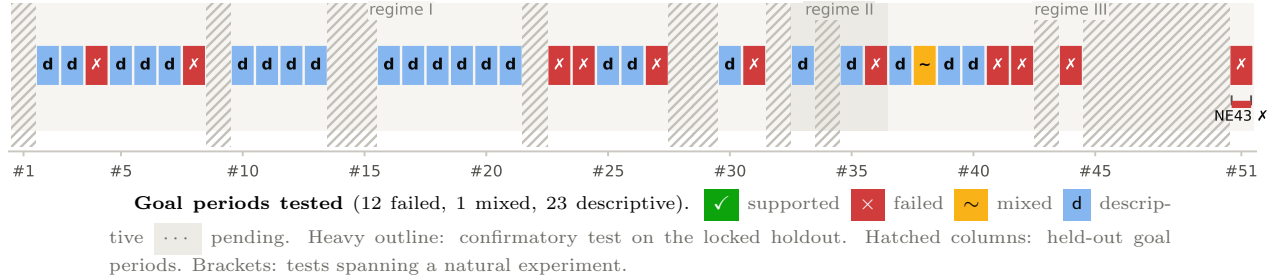
H43: Kicks leave a refractory window

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. In #51, one read is one kick: a second directed item in the same call adds 0.11 [0.01, 0.21] of the first, robust to measured starts and start bounds.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



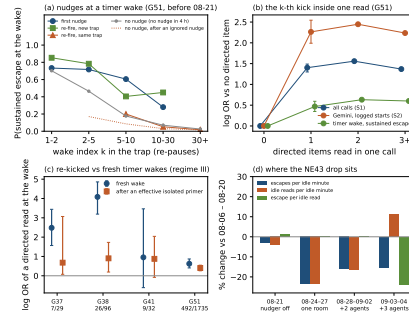
WHAT HAPPENED

H43 asked whether a kicked agent ignores a second kick until the task the first kick launched is over. Each kick is timed at the model call that first reads it (DQ1 ledger). Round 1 (35 non-reserved periods) found no refractory window: a second kick read by a later call keeps 80–130% of a first kick’s effect. Round 2 asks what the unit of operator attention is. Four tests ran, each validated first on synthetic outcomes over the real skeleton: the k -th kick inside one read, nudge facilitation against the nudger’s own trigger, re-kicked timer wakes in every regime-III period, and the decomposition of the NE43 drop. In #51 one read is one kick. A second directed item in the same call adds $\rho_2 = 0.11$ [0.01, 0.21] of the first; a kick read by the next call keeps about 0.7.

PHYSICAL MODEL

Model 09 (renewal Hawkes with a recovery function) and model 03 (refractory compartment). The read-out call is the clock. At each receiving call, the log odds of talking (or of a sustained escape at a timer wake) carry an agent intercept, dose terms f_d for d directed items read in that call, and nuisance terms. These are the read window (H50’s length bias), undirected items, wake index k and trap age (H16), and hour of the day. Rivals: additive kicks ($\rho_2 = 1$), nudge facilitation, and selection by the nudger’s trigger.

$$\begin{aligned} \text{logit } P(y_c) &= \alpha_a + \sum_d f_d \mathbb{K}[\text{dose}_c = d] + \beta_w \ln w_c + \beta \cdot \mathbf{x}_c \\ m_k &= f_k - f_{k-1}, \quad \rho_2 = m_2/m_1 \quad (\text{additive: } \rho_2 = 1) \\ \Delta_F &= \gamma_{\text{refire}} - \gamma_{\text{first}} \text{ at matched } (k, \text{ trap age, prior-nudge state}) \\ \ln R &= \ln \frac{G}{M} + \ln p \quad (\text{escapes} = \text{reads} \times \text{escape per read}) \end{aligned}$$



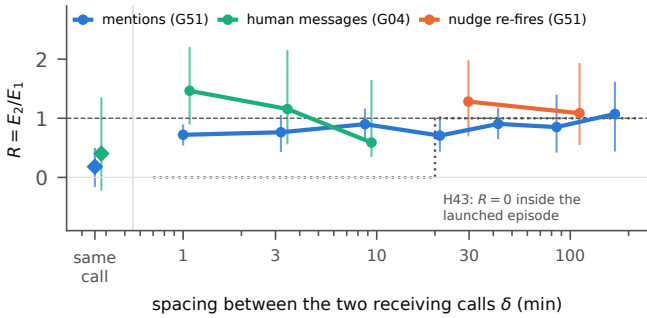
Observables. (a) Sustained escape at a timer wake by wake index (#51, before 08-21). First nudges and re-fires after a sustained run lift escape strongly. A re-fire into the same unbroken trap does almost nothing. (b) Dose curve in one read: the second item adds little and the third nothing, for all calls and for Gemini calls with measured starts. (c) Directed read at fresh vs re-kicked timer wakes per period (small periods: post hoc prior). (d) The NE43 drop sits in the one-room week and in the read cadence, not on the nudger-off day.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not stack kicks before the recipient’s next model call. In #51 a second directed message in the same read adds about a tenth of the first, and a third adds nothing. Merge them, or send the second after the next call; a kick read by the next call keeps about 70%. Do not re-nudge an agent that ignored a nudge and is still in the same pause chain. Its escape rises only from 0.03 to 0.05. Wait until it has worked and stopped again; then a nudge works like a first one. Before blaming a lever for a swarm-level drop, split the drop into read cadence and escape per read. The NE43 drop was longer pauses.

RESULTS VS PREDICTIONS

Prediction (round 2)	Observed	
R3-P1 $\rho_2 \leq 0.5$ (G51)	0.11 [0.01, 0.21]; $m_3 = -0.19$	✓
R3-P3 provider, timing	Gemini 0.08; bounds 0.08 / 0.30	✓
R3-P4 pooled, $I^2 < 0.5$	$\rho_2 = -0.15$, I^2 0.95 (31 periods)	~
R3-P5 next read $R \geq 0.7$	Gemini 0.68 [0.26, 1.14]	~
R2-P1 no facilitation	$\Delta_F = -0.06$ [-0.84, 0.61] (power 0.5)	?
R2-P2/P3 proxies, agent-day	shifts 0.14, 0.18	✓
R5-P1 wake $R_w \geq 0.7$ (G51)	0.64 [0.38, 1.08]	~
R5-P3/P4 homogeneous; G38 untestable	I^2 0.93; G38 R_w 0.22	×
R6-P1 per-read escape $\geq 50\%$	cadence 69%	×
R6-P2/P3/P4/P5 roster, channels, bookends, date	newcomers +24%; nudger 4%; 08-21 flat	✓



Second figure. Round 1. Second-kick effect relative to a first kick against the spacing of the two receiving calls. Diamonds: the second kick read in the same call. Mentions (G51) dip only to ~ 0.8 (15 periods: 0.78). Human messages (G04) and nudge re-fires (G51) stay at or above 1. Dotted: H43's prediction (a 20-min window).

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 × failed
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 × failed
G10 d descriptive
G11 d descriptive
G12 d descriptive
+26 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The batching ratio is a #51 result; elsewhere, talking at the call does not carry the read-out (I^2 0.95). R2 has power 0.5, so neither facilitation nor its absence is shown. A trap that survived a first nudge is selected for non-response, so the same-trap loss is not identified as refractoriness. The large first-nudge effect at the wake ($\ln OR$ 1.0–1.7) is post hoc and model-dependent; it disagrees with round 1 and with H35's minute-level result. Small regime-III periods separate and need a post hoc prior. The NE43 decomposition is an accounting: what links the room merge to longer pauses is not identified.

ROUND 2

Confirm one-read-one-kick on the #51 tail (C2, extended to ρ_2). Separate same-trap refractoriness from trap selection with the NE41 forced erasure between two nudges. Find what lengthened pauses after 08-24. Re-run the first-nudge wake effect against H35's design.

- **Essence.** the unit of operator attention. Round 1 says it is the recipient's read-out (one model call), not its task episode; round 2 should confirm that on the holdout and price the k-th kick inside one read.
- **H43-R1** Run the confirmatory script.
- **H43-R2** Nudge facilitation or nudger selection?
- **H43-R3** Read-out batching as the object.
- **H43-R4** Content-defined episodes and a content outcome.
- **H43-R5** Pooled gate test.
- **H43-R6** Decompose the NE43 swarm drop.

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